**REGRA DA CADEIA** (derivada da função composta)

Exemplo

$f(x) = (x^2 -2)^2$

$f'(x) =2(x^2 -2).2x$

$f(x) =(u(x))^2$;  $u(x) = x^2 -2$  $u'(x) = 2x$

$f'(x) = 2\, u(x)\, u'(x) = 2(x^2 -2)\,(2x) = 4\, x^3 -8x$

$f(x) = (x^3 -2x)^5$

$f'(x)= 5(x^3 -2x)^4.\ (3\, x^2 -2\ )$

$f'(x)= (x^3 -2x)^4.\ (15\, x^2 -10\ )$

*Derivada da função composta (Regra da Cadeia):*

Sejam duas funções diferenciáveis $f$ e $u$, onde $f = f(u)$ e $u = u(x)$, e tal que $y = f\,(u(x))$. Então, $\dfrac{dy}{dx} = f\,'(u)\ u'(x)$ .

**a)** $f(x) = \text{sen}(2x)$.

Seja $u = 2x$. Então $f(u) = \text{sen}\,(u)$ e $f'(x) = f'(u)\,u'(x) = \cos(u)\,2$, ou seja, $f'(x) = 2\cos(2x)$.

**b)** $f(x) = \cos(x^2 + 2x - 1) - 3\text{sen}(x)$.

Seja $u = x^2 + 2x - 1$ e $g(u) = \cos(u)$. Então $g'(x) = \cos'(u)\,u'(x) = -sen(u)\,(2x+2)$.

Assim, $g'(x) = -(2x+2)\,sen(x^2 + 2x - 1)$ e, portanto,

$f'(x) = -(2x+2)\,sen(x^2 + 2x - 1) - 3\cos(x)$.

**c)** $f(x) = (2x - 1)^3$

Seja $u = 2x - 1$. Então $f(u) = u^3$ e $f'(x) = f'(u)\,u'(x) = 3\,u^2\,2$. Portanto, $f'(x) = 6\,(2x-1)^2$.

Regra da cadeia

$f = f(u)$

$u = u(x)$

$y = f(u(x)) = (f_0 u)(x)$

**$y'(x) = f'(u)u'(x)$**

d) $f(x) = e^{3x}$

$u(x) = 3x$

$f = f(u) = e^u$

$f'(x) = f'(u)u'(x) = e^u 3 = 3\,e^{3x}$

$f(x) = e^{5x+2}$ ;

$f'(x) = 5e^{5x+2}$

e) $f(x) = \ln(x)$

$f'(x) = 1/x$

f) $f(x) = \ln(x^3+2x)$

$f(x) = \ln(u(x))$ ,  $u(x) = x^3+2x$  $f(u) = \ln(u)$

$f'(x) = \frac{1}{u(x)} \, . u'(x) = \frac{3x^2+2}{x^3+2x}$

h) $f(x) = sen^2(x) = (sen(x))^2$

$f = f(u) = u^2$

$u = sen(x)$

$f'(x) = 2u.\cos(x)=2sen(x)\cos(x)$

i) $f(x) = sen(x^2)$  $u = x^2$  $f(u) = sen(u)$

$f'(x) = \cos(u).2x = \cos(x^2).2x$